HDOC-56

First Negative Integer in Every Sliding Window

Problem Statement

Write a C program that reads the size and elements of a variable-length integer array arr[] and an integer k such that 0 < k <= size. Pass the array to a function first_negative() using its address. Using pointers and a nested-loop sliding-window approach, display the first negative integer in each contiguous subarray of size k. Display 0 if a window has no negative integer, and display -1 if k is invalid.

Algorithm / Logic

1. Read the array size and elements, then read k.
2. Call first_negative(arr, size, k), passing the array address.
3. If k <= 0 or k > size, print -1 and return.
4. For each window starting from index i = 0 to size - k, use an inner loop from j = 0 to k - 1.
5. Access each element with pointer arithmetic, *(arr + i + j). Print the first negative value found and stop checking that window.
6. If no negative value is found in the window, print 0. Separate all results by spaces.

C Program

```
#include <stdio.h>

void first_negative(int *arr, int size, int k)
{
    int i, j, found;

    if (k <= 0 || k > size) {
        printf("-1\n");
        return;
    }

    for (i = 0; i <= size - k; i++) {
        found = 0;

        for (j = 0; j < k; j++) {
            if (*(arr + i + j) < 0) {
                printf("%d", *(arr + i + j));
                found = 1;
                break;
            }
        }

        if (!found)
            printf("0");

        if (i < size - k)
            printf(" ");
    }
    printf("\n");
}

int main(void)
{
    int size, k, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    if (size <= 0) {
        printf("Invalid array size.\n");
        return 0;
    }

    int arr[size];

    printf("Enter %d elements: ", size);
    for (i = 0; i < size; i++)
        scanf("%d", &arr[i]);
```

```
    printf("Enter k: ");
    scanf("%d", &k);

    printf("First negative integers: ");
    first_negative(arr, size, k);

    return 0;
}
```

Test Cases

S.No.	Size	Array Elements	k	Expected Output	Remark
1	8	12 -1 -7 8 -15 30 16 28	3	-1 -1 -7 -15 -15 0	Valid mixed array
2	5	1 2 3 4 5	2	0 0 0 0	No negative values
3	6	-5 2 -3 4 5 -1	1	-5 0 -3 0 0 -1	Window size 1
4	4	2 -6 7 8	5	-1	Invalid k

Sample Output

```
Enter size of array: 8
Enter 8 elements: 12 -1 -7 8 -15 30 16 28
Enter k: 3
First negative integers: -1 -1 -7 -15 -15 0
```

Result

The program successfully finds and displays the first negative integer in every contiguous window of size k using pointers and nested loops. It prints 0 when a window contains no negative integer and -1 when k is invalid.